$BC = 5 =$ radius.

So, the equation of the required circle is $(x-2)^2+(y-1)^2 = 5^2$

$$x^2+y^2-4x-2y-20 = 0.$$

Example 5.8

A circle of radius 3 units touches both the axes. Find the equations of all possible circles formed in the general form.

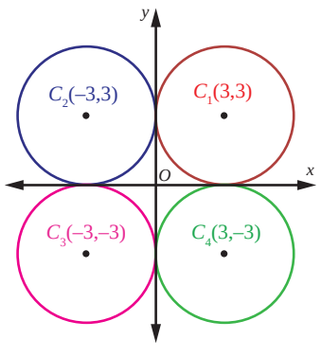


Fig.5.12

Solution

As the circle touches both the axes, the distance of the centre from both the axes is 3 units, centre can be $(\pm 3, \pm 3)$ and hence there are four circles with radius 3, and the required equations of the four circles are $x^2+y^2 \pm 6x \pm 6y+9=0$.

Example 5.9

Find the centre and radius of the circle $3x^2+(a+1)y^2+6x-9y+a+4=0$.

Solution

Coefficient of x^2 = Coefficient of y^2 (characteristic (ii) for a second degree equation to represent a circle).

That is, $3 = a+1$ and $a = 2$.

Therefore, the equation of the circle is

$$3x^2+3y^2+6x-9y+6 = 0$$

$$x^2+y^2+2x-3y+2 = 0$$

So, centre is $\left(-1, \frac{3}{2}\right)$ and radius $r = \sqrt{1+\frac{9}{4}-2} = \frac{\sqrt{5}}{2}$.

Example 5.10

Find the equation of the circle passing through the points $(1,1)$, $(2,-1)$, and $(3,2)$.

Solution

Let the general equation of the circle be

$$x^2+y^2+2gx+2fy+c = 0 \quad \ldots (1)$$

It passes through points $(1,1), (2,-1)$ and $(3,2)$.

Therefore, $2g+2f+c = -2, \quad \ldots (2)$

$4g-2f+c = -5, \quad \ldots (3)$

$6g+4f+c = -13. \quad \ldots (4)$

(2) – (3) gives $-2g+4f = 3. \quad \ldots (5)$

(4) – (3) gives $2g+6f = -8. \quad \ldots (6)$

(5) + (6) gives $f = -\frac{1}{2}$

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